AGENTIC INTERACTION INTELLIGENCE

D–J–O THINK

Making Student Thinking Visible in AI-Assisted Learning

Draft · Judge · Own Architecture & Live Demo

THE ASSESSMENT CRISIS

Beyond the Polished AI Answer

TRADITIONAL PARADIGM

The Output Trap

Deliverable Focus: Essays and exams test completed answers.

Illusion of Mastery: Fluent LLM text masks surface-level learning.

Detector Failure: Probabilistic AI checkers produce high error rates.

D–J–O THINK PARADIGM

The Process Shift

Collaborative Journey: Documenting how students engage with AI.

Visible Inquiry: Tracking questions, challenges, and citations.

Silent Analytics: Non-intrusive observation without auto-grading.

STUDENT JOURNEY · PART 1

Independent Draft & AI Workspace

STEP 1 · PRE-AI DRAFT

DRAFT

Independent Cognitive Baseline

Students articulate understanding before accessing AI assistance.

"Antimycin A blocks Complex III, so electrons cannot pass from ubiquinol to cytochrome c..."

STEP 2 · AI WORKSPACE

JUDGE

Uninterrupted Chat Flow

Students challenge AI explanations; responses serve as context only.

"YOU: But Complex II transfers to ubiquinone. Why would oxygen consumption decrease?"

STUDENT JOURNEY · PART 2

Final Answer & Reflective Ownership

STEP 3 · FINAL ANSWER

OWN

Synthesized Explanation

Final integrated reasoning linked to supporting course materials.

"Source: Biochemistry lecture notes — Mitochondrial electron transport chain."

STEP 4 · REFLECTION

REFLECT

Metacognitive Justification

Documenting whether AI claims were accepted, rejected, or modified.

"Decision: Modified the explanation. I realised both entry pathways converge at ubiquinone."

ARCHITECTURE

Two-Layer Multi-Agent Execution

LAYER 1 · SYNCHRONOUS CAPTURE

Agent 1 (Evidence Capture): Records events in real-time with zero chatbot latency.

Active During Chat

AGENT 2

Independent Thinking

Pre-AI baseline schema.

AGENT 3

Critical Questioning

Inquiry depth tracking.

AGENT 4

Evaluation & Verify

3-tier claim grounding.

AGENT 5

Judgement & Decision

Modification rationale.

AGENT 6

Intellectual Ownership

Draft-to-final synthesis.

AGENT 7 · AUDITOR

Evidence Validation

Adversarial check & veto.

AGENT 8

Assessment Synthesis

Maps validated evidence to Bloom cognitive levels and competencies.

QUALITY ASSURANCE

Agent 7: The Anti-Hallucination Firewall

Adversarial Interrogation

Detects Unsupported Leaps: Flags when upstream agents infer unproven internal thought.

Stops Double-Counting: Prevents one prompt from scoring across multiple rubrics.

Enforces Abstention: Labels missing evidence as unobserved rather than penalizing.

Three-Tier Verification

Level 1 · Reported
Student claims to have consulted lecture slides.

Level 2 · Documented
Student provides exact citation or passage.

Level 3 · Authenticated
System verifies source directly confirms the claim.

SIGNATURE FEATURE

Cognitive Development & Contribution Map

BASELINE
Pre-AI Draft
Prior Knowledge

→

INTERACTION
AI Proposition
Convergence Point

→

TRANSFORMATION
Student Decision
Verified Challenge

→

SYNTHESIS
Final Answer
Demonstrated Ownership

60-SECOND DIAGNOSTIC
Rapid in-platform sorting checks for unverified conceptual shifts.

TARGETED MICRO-VIVA
Flags high-divergence cases (top 10–15%) for brief 3-minute oral reviews.

THE CORE THESIS

AI platforms generate answers. D-J-O THINK documents how students work with those answers.

Think first. Collaborate critically. Own your learning.

Thank You · Open for Questions & Discussion